STANDARD OPERATING PROCEDURE OF pH for an Aqueous Extract VF-E20

Lab /SOP/ 29



1.Mechanical shaker

2.Glass-wares:Flasks,Beakers,Glass stirrer

3.Thermometer, capable of measuring to 0.5 o C



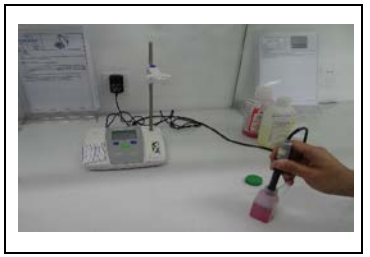
4.Electrode / A device for measuring pH in the range of 0 – 14 to the nearest 0.01 pH unit.

5.Balance nearest 1 mg



Test three (3) samples per material

Cut the laboratory sample into pieces having approximately 2. 5 mm x 2.5 mm.



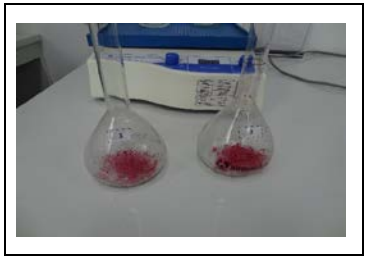
Calibrate the pH meter



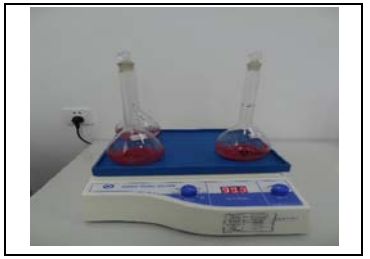
Use the balance to measure the mass of the test material. The mass should greater than 3 g but less than or equal to 5 g.



Use the thermometer to measure the temperature of the water.



Place the test specimen in a flask and add the volume of water.Stopper the flask and shake it vigorously by hand for 30 ± 5 seconds.



Let the flask stand at the standard laboratory conditions with occasional agitation for a period of 6.0 ± 0.2 hours.



Decant the first extract into a beaker, immediately immerse the electrode to a depth to at least 10 mm and stir gently with a rod until the indicated pH-value stabilizes.



Decant the second extract into another beaker, immediately immerse the electrode, without washing, in the beaker to a depth of at least 10 mm and allow to stand without stirring until the pH –value stabilizes. Record this value.



Decant the third extract into another beaker, immediately immerse the electrode, without washing, in the beaker to a depth of at least 10 mm and allow to stand without stirring until the pH –value stabilizes. Record this value.



Record the pH-values of the second and third extracts as the first and second measurements.

Prepared by	Approved by	REVISION		DATE	